

Simone Colli

FREELANCE DEVELOPER & PHOTOGRAPHER | M.Sc. IN COMPUTER SCIENCE

✉ info.dev@simonecolli.com | 📧 simonecolli | 🌐 Simone Colli

Education

University of Parma

MASTER'S DEGREE IN COMPUTER SCIENCE, GRADE: 110/110 CUM LAUDE

Thesis title: Enhancing computational pangenomics workflows. Advisor Prof. V. Bonnici

[Parma, Italy](#)

Sept. 2024 - July 2026

University of Parma

BACHELOR'S DEGREE IN COMPUTER SCIENCE, GRADE: 108/110

Thesis title: A parallel algorithm for computing sequence homology in pangenomic analysis. Advisor Prof. V. Bonnici

[Parma, Italy](#)

Sept. 2021 - July 2024

havce CTF team, University of Parma

CYBERSECURITY COURSE

A cybersecurity training program for university and high school students organized by the Cybersecurity National Lab. (CyberChallenge.it)

[Parma, Italy](#)

2021 & 2024

Artistic roller skater

HIGH-LEVEL ATHLETE

I believe that sports, especially when practiced at a high level, can be considered a means of teaching; not purely under a professional perspective, but a personal one, because it teaches the values of organization, perseverance, patience, resilience, and many other skills that fall under the category of soft skills.

[Italy](#)

2007 - 2023

Experience

Self-employed

FREELANCE SOFTWARE DEVELOPER

- Designed and developed custom software solutions for private clients and small businesses, managing the full software development life cycle.
- Developed web applications and automation tools to streamline business processes.
- Managed client relationships, including requirement analysis, project estimation, and post-release maintenance.

[Remote](#)

January 2026 - Present

Self-employed

FREELANCE PHOTOGRAPHER

Photography services for individuals and events. To handle photo delivery to clients, I independently designed and developed the *Photography delivery* platform (see the Projects section).

[January 2026 - Present](#)

unipr-org

FOUNDER & MAINTAINER

An organization that brings together University of Parma students to collaborate on open-source educational materials and projects. (github.com/unipr-org)

[January 2024 - Present](#)

Tutor

UNIVERSITY OF PARMA

Tutor of the "Sistemi operativi" bachelor course in the Computer Science degree.

[Parma, Italy](#)

February 2026 - June 2026

Freelance Tutor

IN-PERSON & ONLINE

Provided tutoring services for university and high school students, Topics: C, C++, Java, Python, algorithms and data structures, web development.

[February 2022 - January 2026](#)

Tutor

UNIVERSITY OF PARMA

Tutor of the "Ingegneria del Software" bachelor course in the Computer Science degree.

[Parma, Italy](#)

September 2025 - December 2025

Workshop: Use of Graphics Libraries for Software Implementation

UNIVERSITY OF PARMA

Intensive bootcamp focused on the fundamentals of Python programming and the use of a graphics library. The learning path culminated in programming an application (the Pong game), providing participants with practical software development skills.

[Parma, Italy](#)

June 16, 2025 - June 19, 2025

Tutor

UNIVERSITY OF PARMA

Tutor of the "Sistemi operativi" bachelor course in the Computer Science degree.

[Parma, Italy](#)

February 2025 - June 2025

Workshop: Tutoring from Different Perspectives and Supporting IT Tools

Parma, Italy

UNIVERSITY OF PARMA

January 24 2025

Presented and discussed the various perspectives (student, tutor, and lecturer) and related challenges concerning the role of a tutor

Tutor

Parma, Italy

UNIVERSITY OF PARMA

September 2024 - December 2024

Tutor of the "Ingegneria del Software" bachelor course in the Computer Science degree.

Internship

UNIVERSITY OF PARMA

February 2024 - June 2024

University internship focused on the development of parallel software for pan-genomic analysis. Languages used: C++, Python, Bash.

Projects

MGP — Production management

WORK (CONTINUOUSLY EXPANDING)

- Full-stack web platform for managing a manufacturing company's production department: product configuration, automatic generation of bills of materials and documents (PDF/DXF), and truck load-plan optimization.
- Polyglot microservices architecture, documented with ADRs and C4 diagrams. Stack: Laravel (PHP), React / TypeScript, Python (FastAPI/Flask), MySQL, Docker, Kubernetes (k3s).
- Industrial OT/IT integration: communication with factory PLCs via OPC-UA, constraint-based load-plan optimization with MiniZinc and Google OR-Tools.

Product configurator

WORK (IN DEVELOPMENT)

- Web-based 2D product configurator: a graph-based guided question-flow engine for client-side configuration and a canvas for composing components with DXF export.
- Stack: Laravel 12 / PHP 8.4, React 19 / TypeScript, PostgreSQL (domain modeled on jsonb columns).
- Developed with Test-Driven Development, an ADR-driven architecture and a security-by-design approach (server-side validation, anti-DoS bounds).

Photography delivery

PERSONAL (PROTOTYPE, IN ACTIVE TESTING)

- Self-hosted platform supporting freelance photography work: private galleries protected by an access code, cart, order management and full-resolution file delivery after payment.
- Stack: Python (FastAPI), Nuxt 3 / Vue, PostgreSQL, Redis, RQ (async workers), Docker.
- Strong focus on security and GDPR compliance (AES-256-GCM encryption of billing data, Argon2id, audit log) and a complete quality pipeline: multi-tool security scanning and strict type-checking.

Freelance hub

PERSONAL

- Self-hosted automation and CRM platform for freelance work: manages the full client lifecycle (leads, quotes, projects, time tracking, contracts, payments) with business analytics and workload planning.
- Microservices architecture with Gmail/Google Calendar/webhook integrations and an AI assistant on local LLMs. Stack: Python (FastAPI), GraphQL (Strawberry), Nuxt 3 / Vue, PostgreSQL, MiniZinc, CLI.
- Engineered with 100% test coverage (800+ tests), strict type-checking (mypy/TypeScript) and a complete DevSecOps pipeline (SAST, SBOM, multi-tool security scanning, SonarQube).

Personal website & portfolio

PERSONAL

- Bilingual (IT/EN) personal portfolio website showcasing projects, skills, talks and a dedicated photography section.
- Static site generation (SSG) via pre-rendering for SEO and performance. Stack: React 19, TypeScript, Tailwind CSS 4, Vite.
- Internationalization with i18next, curated SEO (Open Graph, Twitter Card, canonical), Google Analytics 4, and continuous automated deployment to GitHub Pages via GitHub Actions.

Tools and Softwares

Quantum Portfolio Optimization

PYTHON

2025

Development of a Python-based quantum portfolio optimization tool, leveraging the power of quantum computing to enhance portfolio management and investment strategies. The project explores the potential of hybrid algorithms VQE e QAOA, that combine classical and quantum computing, to solve Quadratic Unconstrained Binary Optimization (QUBO) problems. Simulations have been designed and implemented using Qiskit to evaluate algorithm performance under noiseless and noisy conditions.

(github.com/simonecolli/quantum-portfolio-optimization)

Ottimizzazioni garanti accademici

ASP, PYTHON 2024
Development of an automated system for assigning academic guarantors to university courses, adhering to complex ministerial regulations. The project includes the creation of a data preprocessing pipeline and the implementation of a system based on Answer Set Programming (ASP) for dynamic optimization of resource allocation. The system ensures scalability across various datasets, reduces reliance on external staff, and meets institutional and regulatory requirements. (github.com/simonecolli/ottimizzazione-garanti-accademici)

PanDelos-plus

C++, PYTHON, BASH February 2024 - June 2024
A C++ based, parallelized revision of PanDelos (github.com/InfOmics/PanDelos) software for pan-genome content discovery. (github.com/simonecolli/PanDelos-plus)

Deep Neural Network Library

C++, PYTHON From May 2024
A flexible, modular, easy-to-use, and high-performance C++ library for building, training, and using deep neural networks. (github.com/unipr-org/deep-neural-network)

MazeProject

HTML, CSS, JS From July 2022
A user-friendly web-based application developed for the European Research's Night, aimed at making the understanding of graph-based search algorithms more interesting, easier, and enjoyable. (github.com/simonecolli/MazeProject)

Certificates

2022 **Responsive Web Design**, FreeCodeCamp
now **Data Visualization**, FreeCodeCamp

Conferences

BBCC-Meetings Napoli
BBCC2024 27-29 novembre 2024
Oral presentation of PanDelos-plus.

Technical skills

Main computer languages	C++, Java, Python, TypeScript, PHP, Bash scripting, HTML, JavaScript, CSS, SASS
Other computer languages	C, C#, Objective-C, LaTeX, Markdown, Matlab, YAML.
Operative systems	Linux, Windows, MAC OSX.
Environments	Visual Studio Code, Eclipse, Android Studio, Xcode.
Databases	PostgreSQL, MySQL, Redis.
Creative instruments	Microsoft office, Darktable, GIMP, Adobe Illustrator, Adobe Lightroom Classic, Audacity.
Frameworks and other tools	React, Vue.js, Nuxt, FastAPI, Flask, Laravel, GraphQL, Tailwind CSS, jQuery, Bootstrap, Vite, GitHub Actions, Grafana, Docker, Kubernetes, git, nginx, Apache Web Server, Oracle Virtual Box, Masterwood.
Soft skills	Organization, problem solving, teamwork, stress management, self-development, resilience, flexibility, patience, punctuality, perseverance.
Languages	Italian (First language), English (Intermediate).